

Abstract

The Product Search and Recommendation System is a Java-based application developed to provide efficient product searching and relevant product recommendations. The system uses a corpus of 300 text files containing product information.

The project implements several Data Structures and Algorithms (DSA) concepts. The Knuth–Morris–Pratt (KMP) and Rabin–Karp algorithms are used for pattern matching. A Trie data structure supports prefix-based searching, while the Levenshtein distance algorithm helps identify and correct spelling mistakes in user queries. The Edmonds–Karp maximum-flow algorithm is used in the recommendation module to select products while satisfying defined recommendation constraints.

The application provides a console-based interface through which users can search for products, explore prefix matches, receive suggestions for misspelled queries, and view product recommendations. It also includes a benchmarking command to compare the execution times of the KMP and Rabin–Karp algorithms.

The main objective of this project is to demonstrate the practical application of data structures, string-matching algorithms, dynamic programming, and graph algorithms in a product search and recommendation problem.